Reading Notes for Week 8

WaveNet

- A fully probabilistic and autoregressive model based on PixelCNN designed for generating sequential outputs.
    - Output of the model has the same time dimensionality as the input.
    - No pooling layers in the network.
    - Output variable is categorical.
- Causal Convolution
    - Like in the RNN model, we cannot have any information about the future prediction, the current state is totally determined by its past.
        - p(xt|x1,x2,...,x_{t-1})
    - Faster than traditional LSTM without memory units.
    - Using a causal convolution layer to mimic the recurrent behaviors of LSTM or other sequential-sequential models.
- Dilation Causal Convolutional Layer
    - Idea: Skip some of the pooling output by specifying a dilation parameter.
    - Model is two-folded. Like a binary tree, can achieve the same receptive field with simplified model structure.
    - Having a big receptive field is important for TTS.
        - If we have 4 layers, we will have 2^4 + bias amount of reception field.
- Gated Activation Units
    - Not the same gating mechanism like in LSTM.

Stochastic Backpropagation(Memory_Efficient_Model_Training)

- Idea: During backward propagation, SBP calculates the gradients by only using a subset of feature maps to save the GPU memory and computational cost.
- SBP saves memory by **only caching a subset of feature maps in the backward pass**, but it **keeps all the full forward paths.**
- An efficient way to implement stochastic gradient descent by performing backpropagation dropout, which leads to considerable memory saving and training process speedup, with a minimal impact on the overall model accuracy.

Diffusion Model

- Diffusion Models are generative models, meaning that they are used to generate data similar to the data on which they are trained.
    - A latent variable model which maps to the latent space using a fixed Markov chain.
    - Forward process/Diffusion Process: From an image to gaussian noise.
    - Reverse process: From gaussian noise back to image.
- Not requiring adversarial training like in GAN.
- A Diffusion Model is trained by finding the reverse Markov transitions that maximize the likelihood of the training data.
    - Training: Minimizing the variational upper bound on the negative log likelihood.

    - The transition distributions in the Markov chain are Gaussian, where the forward process requires a variance schedule, and the reverse process parameters are learned.
- Diffusion Models are highly flexible and allow for any architecture whose input and output dimensionality are the same to be used. Many implementations use U-Net-like architectures.